

MeltyFC – ERRATA PREPASS (knowledge-based seed) · 2026-07-09

**Fable's L1 contribution: known-major silicon
errata for our six parts, dispositioned**

**against the actual tree. EPISTEMIC STATUS:
training-knowledge recall, NOT the current**

**ES PDFs – this is the FLOOR of the errata
audit, never its ceiling. Opus's full**

**ES-document walk (ES0182/F405 ·
ES0287/F411 · F72x + ES0290/F74x ·
ES0392/H743 ·**

**H72x ES) remains MANDATORY and must diff
against this file: confirm every**

**disposition, extend with every erratum this
recall missed.**

Tags: [HIGH] = assert with confidence · [MED]
= verify exact wording in the ES doc.

E-01 · AFFECTED — F4 I2C BUSY/analog-filter lockup [HIGH] · the prepass's headline

The erratum: F405/F411 I2C peripheral can latch BUSY after a glitch or misplaced START/STOP on the lines — and it stays latched through re-initialization. This is the *master/peripheral-side* wedge, distinct from a latched slave.

Our tree: `i2c_recovery.hpp` (verified, full read) handles exactly the SLAVE-side case — 9× SCL + STOP clears a latched H3LIS, then retries `Wire.begin()`. **A**

BUSY-latched master survives all of that: the retry loop reconfigures pins and re-inits, but the erratum's whole point is that re-init doesn't clear it.

Disposition: AFFECTED → fix. Add step 4.5 to the recovery sequence, each retry:

force-reset the I2C peripheral via RCC

(`__HAL_RCC_I2Cx_FORCE_RESET()` → short delay → `RELEASE_RESET` → re-init). Family-portable, three lines, and it's ST's

own documented workaround class. Update the header's step list + the eventual per-family implementation. With both halves, the module cures a latched slave AND a latched master — the complete wedge set.

Bonus note the module surfaced: its steps 5–7 say "Arduino Wire (portable)" — the tree builds under the stm32cube framework. The framework tension flagged in Round 6 touches this module's implementation plan; resolve when the sensor wiring lands.

E-02 · NOT AFFECTED, with cited rationale — F4 DMA2 concurrent-AHB corruption [HIGH]

The erratum: F4 DMA2 can corrupt data when servicing AHB-side targets (mem-to-mem, GPIO) concurrently with APB peripherals.

Our tree (verified): every DMA transfer in every family is `DMA_MEMORY_TO_PERIPH` into APB timers; zero mem-to-mem, zero AHB-target transfers exist (f4/dshot:190, f4/ws2812:66, and the h7 equivalents).

Disposition: NOT AFFECTED — by usage pattern.

Guard the disposition: if anyone ever adds GPIO-DMA or mem-to-mem on DMA2, this

erratum re-opens; one line in
PORTING.md's f4 section.

E-03 · NOT AFFECTED, with cited rationale — F4 DMA FIFO/burst errata family [HIGH]

FIFO/burst corner-case errata require FIFO mode +
burst configs. **Our tree:**

`FIFOMode = DMA_FIFOMODE_DISABLE` **explicitly, every
stream, every family** (f4:197,

h7:140, h7/ws2812:74 — and direct mode is the F7
default). No bursts configured.

**Disposition: NOT AFFECTED — direct mode
everywhere, and it's explicit, not default-
accidental** (the R6-4 discipline paying off in an errata
audit).

E-04 · NOT AFFECTED BY DESIGN — F7 D-cache/DMA coherency errata family [HIGH]

The F7 cache-vs-DMA advisories (stale/corrupt data
on cacheable DMA buffers) are

moot for us: **all F7 DMA buffers live in DTCM, which
is never cached** (dma_buf.h

l-11a, boot-asserted, ldscript-placed, map-gated).

Disposition: architecture immunity,
three enforcement layers, cite and close.

E-05 · NOT AFFECTED, must become DELIBERATE — H743 revision-Y limits [HIGH]

Early H743 (rev Y) caps at 400MHz and lacks VOS0.
Our config is 400MHz/VOS1 —

revision-proof by accident. Disposition: make it
deliberate — (a) the 480MHz

"later experiment" note in target.h gains "rev V+ only,
read DBGMCU_IDC/REV_ID

first"; (b) BENCH_TEST gains "record silicon revision
ID at first connect, per

board." Costs nothing; prevents a future 480 attempt
on Y silicon from becoming a
mystery.

E-06 · WALK-VERIFY set [MED — do not close without the ES docs]

- **F7 new-IP I2C errata** (timing/NOSTRETCH corner cases): our usage is standard master, clock-stretch defaults — likely clean, verify wording.

- **H743 DMAMUX** overrun/sync-event flag semantics: driver should clear-on-configure; confirm against ES + RM notes when the DMAMUX layer is exercised.
- **H743 SDMMC errata** (early-rev advisories): our blackbox plan is simple single-block writes — verify the advisories don't touch single-block, or pin the workaround.
- **H743 RAM/ECC advisories per revision** [MED — recall is fuzzy here; walk the ES doc's RAM section rather than trusting this line either way].
- **H72x (H723/725)**: thin recall on a newer part — the ES walk owns this one nearly wholesale. Flag specifically: SMPS transition/timing notes, and anything touching the VOS0/520MHz region before the parked experiment ever runs.
- **USB OTG FS minor errata** (SOF/session quirks, all families): CDC-only usage is usually untouched; verify.

E-07 · Cross-cutting, non-silicon

- **IWDG hardware-start option byte**: if ever set, the watchdog runs from reset — before init — guaranteeing the boot-loop I-7 exists

to prevent. Provisioning doc
(L7) gains: "verify IWDG_SW = software start" as a
checked field per board.

- **HAL library bugs (software, not silicon):** the
mitigation is P-01 — pin the
platform version, record the resolved HAL version
per phase report, and treat HAL
updates as reviewable changes, not background
drift.

CONTRACT FOR THE FULL ES WALK (Opus)

Fetch each part's current ES document. For EVERY
erratum touching TIM, DMA/DMAMUX,
I2C, SPI, USB, RCC/PWR, FLASH, IWDG, SDMMC: one
table row — erratum ID · affected
rev(s) · our exposure (with file:line or config cite) ·
disposition
(AFFECTED → fix / NOT-AFFECTED → rationale /
DEFERRED → bench gate). Diff against this
prepass: every E-xx above either CONFIRMED or
CORRECTED with the ES text quoted.
Output: docs/ERRATA_AUDIT.md. Fable reviews the
dispositions — it's a

claims-vs-silicon audit, which is by now the house specialty.